

Project 2: Gene Expression Analysis

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1 Background

As *Saccharomyces cerevisiae*, a strain of wine making bacteria, goes through different phases in its metabolic process, different genes vary their level of expression. This means that some genes are likely to be more connected to each other than others, implying that some genes are specific to certain processes.

2 Initial Exploratory Data Analysis

Provided was a dataset described to be the gene expressions of 6,400 genes across 7 different time points.

After importing the dataset I decided to do EDA to clean up and look at the data that was being analyzed. When importing the dataset, I found that there was not 6,400 genes but actually 6153 entries. Of these, only 6102 had unique values for the ORF. Because of this I explored the labeled genes further. I found that there was an ORF that had 2 different names. The gene had the ORF of **YDR272W** but had a name of **GLO2** and an empty name. Both of these 2 entries, and the duplicate ORF entries had different values for the observed data. This raised uncertainty in how to process it, because of this, I dropped all entries that had an ORF duplicated elsewhere.

I also decided to look at the distributions for all the readings to see if it followed any distributions. I looked at three things.

1. Correlations across ratio readings
2. A histogram of the data.
3. Q-Q plots for various distributions

When looking at the correlation between columns, which can be seen in Figure 1, the main observation I saw is that the strongest correlations between columns was the columns directly next to each other. This makes sense generally as the expression levels should be dependent on the change in state between the last time period.



Figure 1: Correlation Heatmap of Ratios

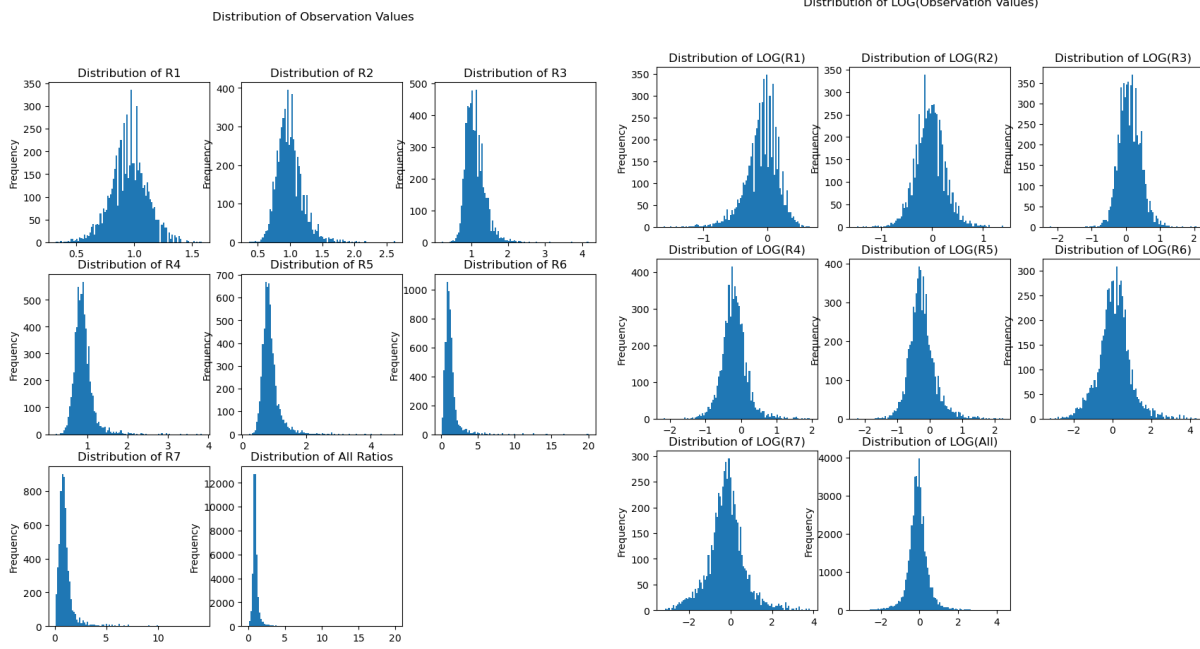


Figure 2: Histogram of Observed Ratios

Figure 3: Distribution of Data after Log Transformation

When looking at the histogram in Figure 2, I noticed that the first observation looked like a normal distribution, and as the time increased, the distributions became more and more right skewed, with R7 being very right skewed.

The next thing I tested was applying a log transformation. Figure 3 shows the distributions after applying the transformation. This made the distribution look more normal, suggesting it follows a log normal distribution.

After seeing this, I made a Q-Q plot to test my assumption. Figure 4 shows that the model is mostly log normal, with a few outliers. These are likely the high variance nodes that are affected by the diaxulic shift.

To test this assumption I plotted another Q-Q plot, with the author identified high variance genes removed. Figure 5 shows a clear increase in the quality of the q-q plot. This indicates that the

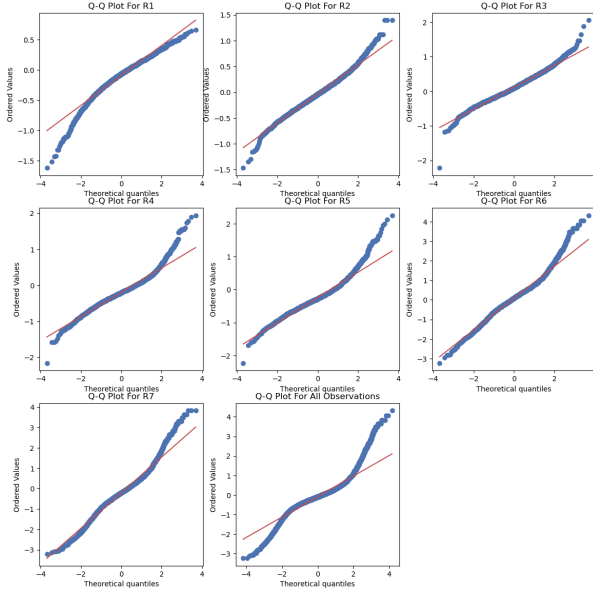


Figure 4: Distribution of data after Log Transformation

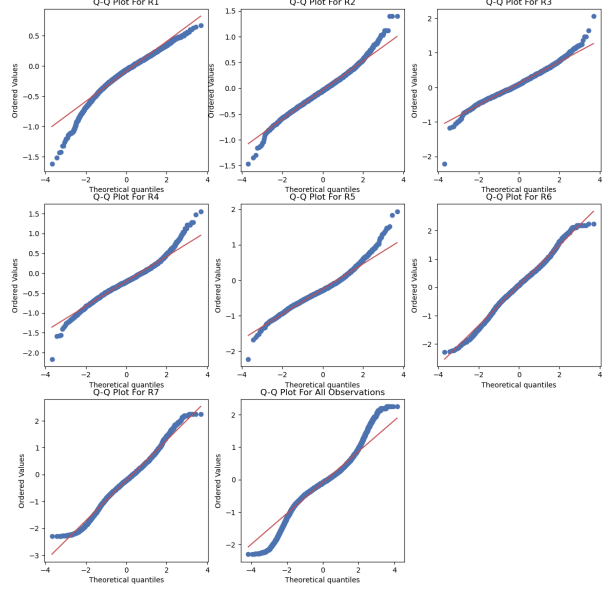


Figure 5: Distribution of data after Log Transformation (Without Author Identified Top 230)

overall background distribution of the data follows a log normal data. Therefore, it is reasonable to conclude that data unlikely to occur in this distribution, may be the result of external factors. In other terms, genes not following the expected distribution, are likely responsible for the resulting behavior.

3 Clustering Analysis

The main goal was to determine which genes were related and to be able to see the functions for each group. The first step in doing this is by clustering the genes together.

3.1 Initial Data

Figure 6 shows a heatmap of the cosine similarities between genes. Each gene was treated as a 7×1 vector where the i th element corresponds to the R_i ratio reading. From here, the cosine of each vector was computed to find the similarities. The assumption is, that related genes will behave similarly to each other.

3.2 Clustering

Before clustering, I preprocessed the data by applying a log transformation on all values due to my analysis in Section 2. This allowed for essentially normal distributions on all ratios, meaning no single ratio reading would bias the cosine similarity of each vector.

When clustering, I tried using two different methods. My first attempt used agglomerative clustering, where a distance threshold determined if two genes could be clustered together. My second attempt was using regular k -means.

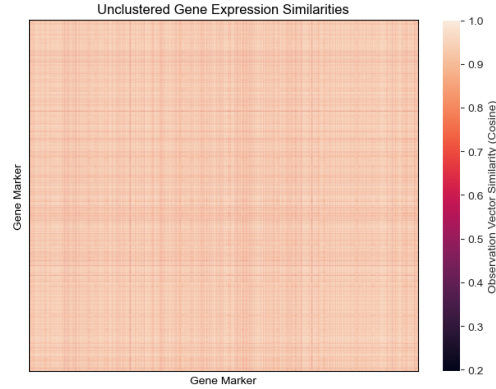


Figure 6: Heatmap of Similarities (Unsorted)

3.2.1 Agglomerative Clustering

In order to form clusters for gene function, my first attempt took every gene and used average linkage to form hierarchical clustering.

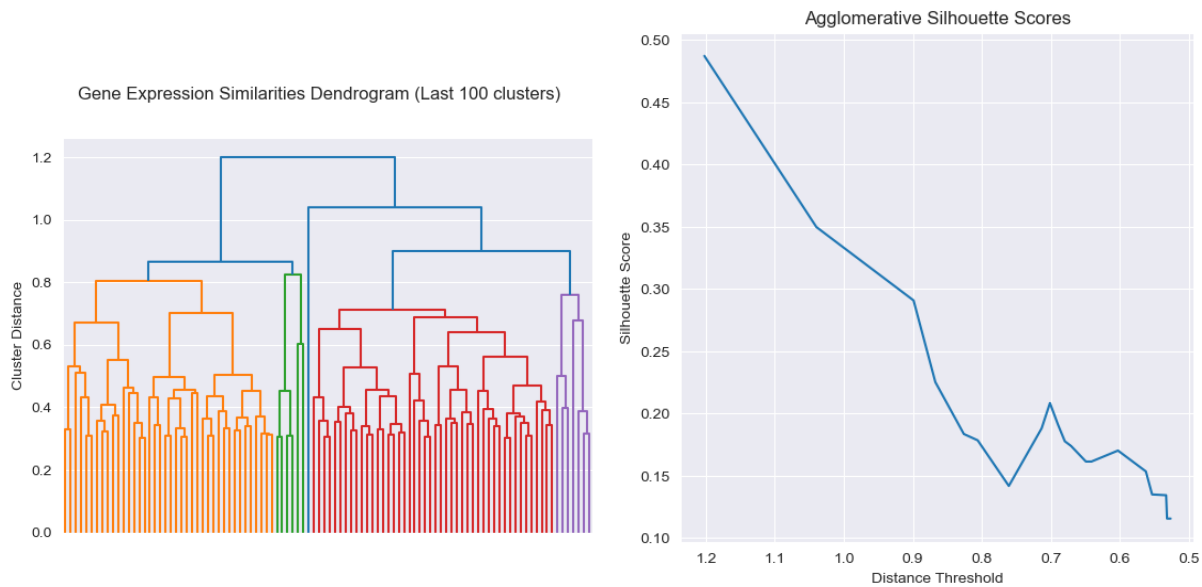


Figure 7: Histogram of Observed Ratios

Figure 8: Distribution of Data after Log Transformation

Figure 7 is a dendrogram showing the distances of each cluster join. Looking at the dendrogram, the best cluster looked like it might be around 0.8. Comparing this to the silhouette graph in Figure 8, shows a similar result of splitting at 0.8.

Figure 9 shows the heatmap sorted with the clusters derived from using a distance threshold of 0.8. This has 5 clusters and a silhouette score of 0.225.

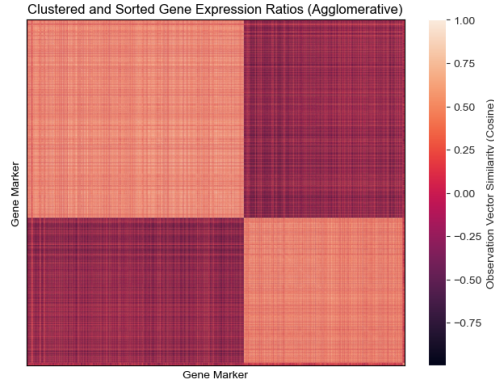


Figure 9: Heatmap of Similarities (Clustered using Agglomerative Clustering)

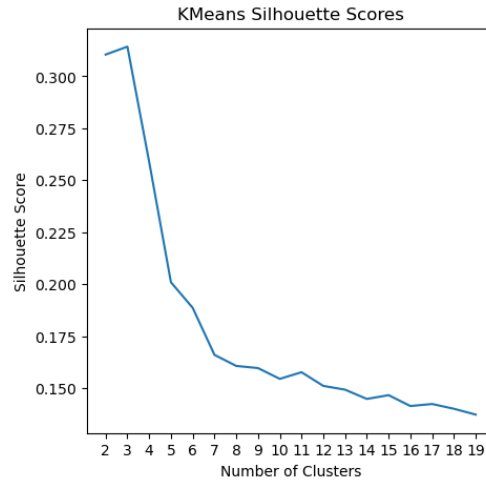


Figure 10: Silhouette Scores of k -means clustering

3.3 k -means

Looking at the results of the agglomerative clustering, the resulting heatmap looked unimpressive. My second approach, used k -means clustering. Because the number of dimensions ($n = 5$) wasn't incredibly large, I clustered on the log transformed observations instead of embeddings.

Figure 10 shows the silhouette scores of k -means clustering on the log transformed data. Using the elbow method, I determined the best values for k would be either 7 or 8.

Choosing $k = 8$ results in silhouette score of 0.24. Figure 11 shows the resulting heatmap of similarities. This figure looks significantly better than the one from agglomerative clustering.

3.4 Conclusion

Based on the dataset, k -means clustering was better at finding the patterns within the data. The next step in analysis, is separating the genes that effect the diaxulic shift from the background noise.

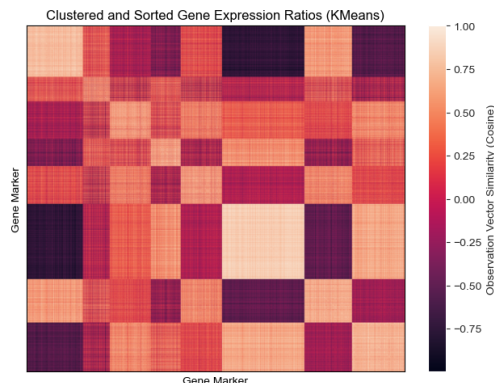


Figure 11: Heatmap of Similarities (Clustered using k -means)

4 Variance Scores

Provided was a list of 230 genes that were identified by the authors as genes with high variance. On doing initial EDA of this set I found that the values of the ratios had a log base 2 transformation on them from the original data. There also wasn't 230 genes, but actually 228 unique ORFs. Of the genes that were initially excluded in Section 2, only 225 were in both lists.

When finding my list of genes that were high variance I could think of a few main methods.

4.1 Observation Probability

My first method assumed that genes that are unlikely to be following the background distribution are responsible for the gene expression. For each time point, I calculated the mean and standard deviation. For each gene observation at the time point, I calculated the two-tailed probability that the value would be observed assuming the distribution is normal. After 7 probabilities were calculated (1 for each time point) for each gene, the joint probability was calculated. The joint probabilities were sorted and the ones with the values were determined to be the important genes. Choosing the top 225 genes, had 164 in common with the author identified top 230. This leaves this method with a Jaccard score of 0.573.

This method overall isn't great. It assumes the most variable genes are those that are unlikely to occur. However, if a gene always has high or always has low expression levels then it will be at one end of the normal distribution. This means that the joint distribution will always be more unlikely than a gene that is always at the middle, but has an outlier observation. This method might be best for detecting outliers as opposed to variable genes.

4.2 Variance / Standard Deviation

The next method I tried was simply using the variance of the gene and sorting them in order. This led to a total of 177 genes out of 225, giving it a Jaccard score of 0.64. This is slightly better, but due to the lack of observation time points, could be improved.

4.2.1 Evaluation of Variance List

A PANTHER overrepresentation test was conducted on the top 230 genes identified in 4.2. This test used a binomial distribution and had a p-value of $4.90 \cdot 10^{-11}$ for having the number of genes associated with metabolic processes. It also had a p-value of $4.61 \cdot 10^{-13}$ for having this number

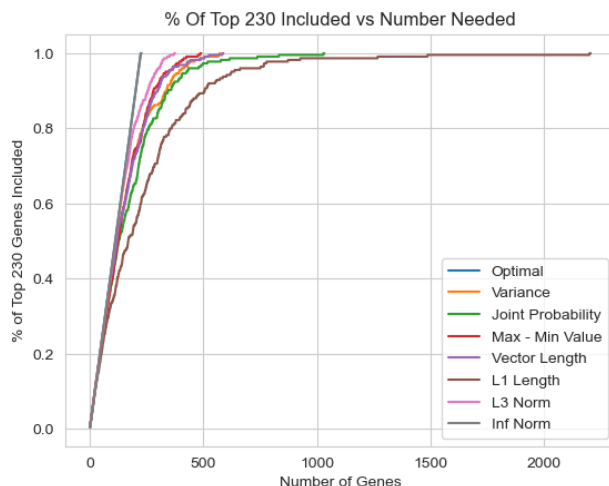


Figure 12: Top 230 Similarity Effectiveness

of genes associated with rRNA metabolic processes. This indicates that this list of genes is very likely to be connected to the metabolic processes of the bacteria.

4.3 Min - Max

My third method involved sorting each gene by the difference between its minimum and maximum expression values. This method led to 172 out of the 225 genes, this led to a Jaccard score of 0.61

4.4 Chasing the Metric

After checking the previous methods, I wanted to find how the authors might have calculated their top 230. Each of my methods involve sorting all the genes and only selecting the top 230. Figure 12 shows how quickly each method contains the top 230. This figure shows that the closest metric is vector length. As the ordinal increases it becomes closer and closer to the author defined metric.

I found a 1-1 match by taking each gene expression vector and sorting them by their infinity norm. This method resulted in a Jaccard score of 1.00.

5 Filtered Clustering

After using the author defined genes with high variance, I reran the clustering analysis.

5.1 EDA

Figure 13 shows the distribution of values from the top 230 genes. Now it can be seen that these observations don't follow a normal distribution like before.

5.2 Clustering

Because the agglomerative clustering was not effective, I focused on k - means clustering. Figure 14 shows the silhouette scores for using k - means clustering. Based on this graph I chose a k value of 6. This had a silhouette score of 0.37. Using $k = 6$ generates the heatmap in Figure 15. This heatmap clearly shows the presence of at least 2 different clusters.

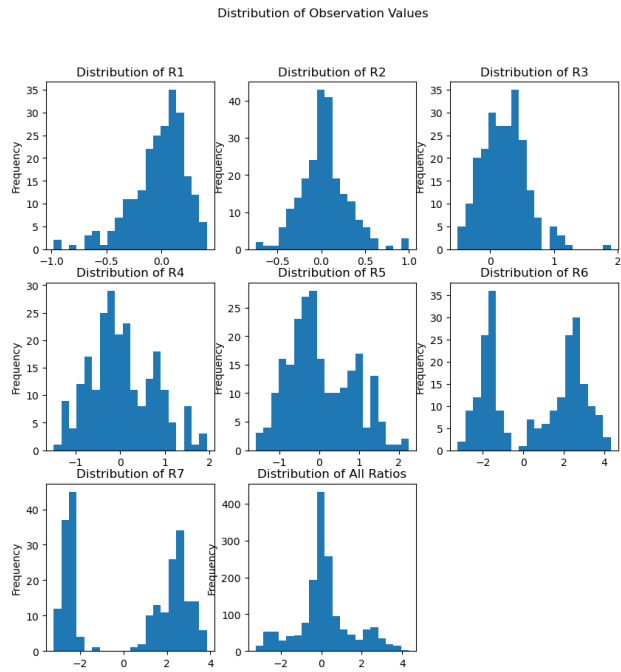


Figure 13: Distribution of Top 230 Genes

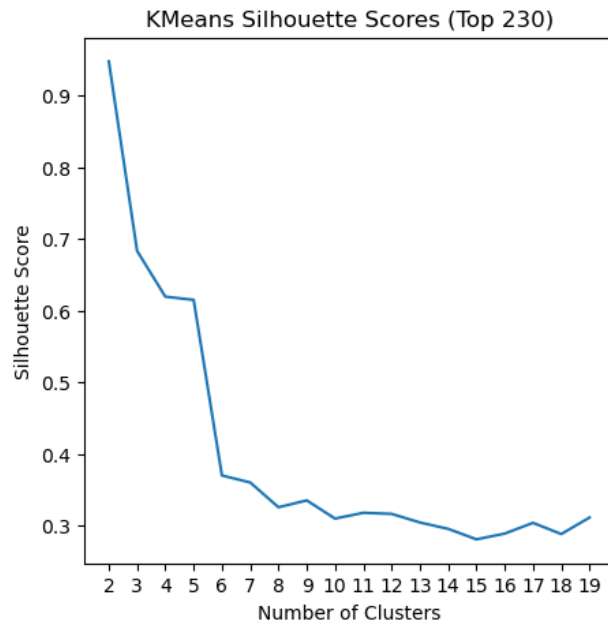


Figure 14: Silhouette Scores of k -means clustering on top 230 genes

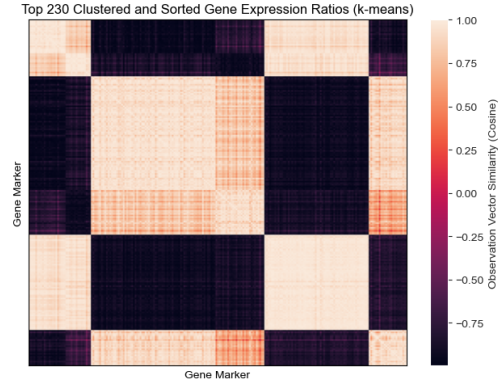


Figure 15: Heatmap of Similarities of top 230 genes (Clustered using k – means Clustering)

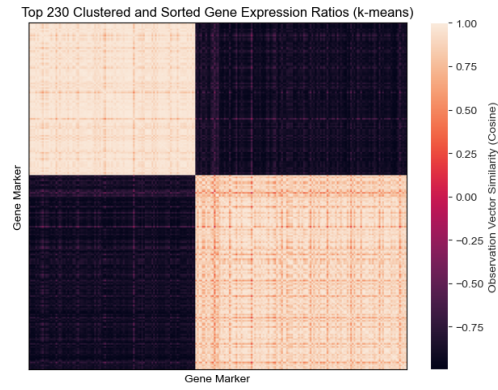


Figure 16: Heatmap of Similarities of top 230 $k = 2$

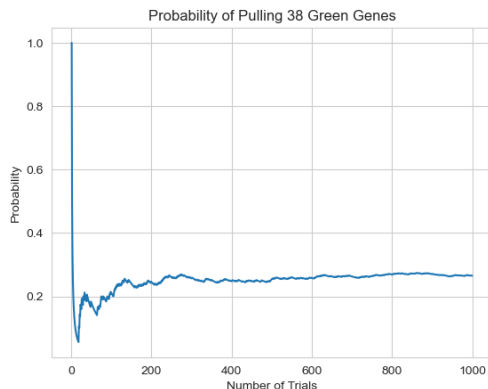


Figure 17: Probability of Drawing 38 out of 50 greens

Visually, the black space forms 2 clusters. Figure 16 shows a similarity matrix graph with only 2 clusters. This clustering has a silhouette score of 0.947.

This shows that the genes that are active at different stages are now visible. The genes in the fermentation stage can be differentiated from those in the inverted metabolism phase.

6 Exploring p -values

The following section goes through various calculations of p – values. For the sake of the exercise suppose that you have a list of 50 items (genes) and 38 of them are colored by (annotated with) *Green*. Of all possible 6000 items, 4260 or 71% are colored with *Green*.

6.1 Calculation via Simulation

After running 10,000 simulations where 50 genes were pulled out of a bag containing 6000 genes (4260 were green), a p -value of about 0.281 was calculated. After running the simulation several times, the value appeared in the range of 0.27 to 0.29, indicating this is a good estimate for the probability of this occurring.

Figure 17 shows the simulated probability over time. By trial 500 the probability seems to have flattened out, indicating that the 1000s trial is a good indicator for the final probability.

6.2 Exact Value

Calculating with the exact value with a binomial function ($k = 38$, $n = 50$, $p = 0.71$) leads the expected value to be 0.2713. This is very close to the trial values.

This means that green is appearing more frequently than expected but not so much to the point that it is unlikely 38 were pulled. This still has a probability of 27% for occurring naturally.

6.3 Probability of Other Values

The probability of pulling less than or equal to 35 green genes is more likely, at 37%. The probability of pulling more than or equal to 40 is 0.10%, which is less likely, but still reasonable for it to have occurred naturally.

6.4 Parameter Search for a Specific p -value

Goal: Find a pair of numbers (n, k) such that if n (number of items in the background) and k (number of items in the research list) are annotated with *Purple*, a student working on this problem will calculate p -value between (0.003 and 0.01) for *Purple*

In order to solve this problem, I assumed that the total number of genes is 6000, n is the number that are purple, and k is the number of genes that are purple out of the 50 that are pulled. The probability of success would then be $\frac{n}{6000}$. My approach used a parameter search where I looked through all values of n in the range $[1, 100]$ and k in the range $[1, 50]$. This approach found 56 values in the range. Some of these included $(1, 1)[p = 0.008]$, $(10, 2)[p = 0.0032]$, and $(89, 4)[p = 0.006]$.

6.5 Significant Overrepresentation

In order for something to significantly overrepresent a group, you need to look at the p -values. In the examples in Section 6.4, these values are significant. If one would were to observe them, they would be a significant overrepresentation of the actual distribution. This is because observing these values is very unlikely.